

MASTER SERVICE AGREEMENT

5-Year Hardware SLA, On-Site Maintenance & Local Spare Parts Support Contract

Contract Identifier: MSA-2026-NEXAGEN-04

Service Provider: NexaGen Systems Global Field Services Inc.

Client Entity: Westford High-Performance Computing Center

Contract Term: 5-Year Continuous Coverage (March 2026 - March 2031)

ARTICLE 1: MASTER SERVICE AGREEMENT (MSA) SCOPE & PURPOSE

ARTICLE I: SCOPE OF MASTER SERVICE AGREEMENT & SYSTEM BOUNDARIES. This Master Service Agreement (MSA) establishes the legally binding framework governing hardware maintenance, technical support, field service dispatch, and software update services provided by NexaGen Computing Solutions Inc. for the Westford HPC Center.

The system boundary encompasses 128 NexaGen X9000 compute enclosures (512 NVIDIA H100 GPUs, 256 AMD EPYC 9654 CPUs), 48 QuantumX800 NDR InfiniBand switches, intelligent rack PDUs, and direct-to-chip liquid cooling manifolds.

This Agreement commences on October 1, 2026, and remains in full force for a primary term of sixty (60) calendar months, unless terminated earlier in strict compliance with Article XII.

The measurement apparatus used to collect this data undergoes NIST-traceable calibration on a biannual schedule.

ARTICLE 2: SERVICE LEVEL AGREEMENT (SLA) TARGETS & UPTIME GUARANTEES

ARTICLE II: HARDWARE AVAILABILITY & UPTIME SERVICE LEVEL COMMITMENTS. Vendor guarantees that the compute cluster hardware shall maintain a cumulative Monthly Hardware Availability Metric of no less than ninety-nine and nine-tenths percent (99.9%).

Hardware Availability is calculated monthly as total operational compute node hours divided by total scheduled operational hours, excluding pre-approved scheduled maintenance windows communicated 14 days in advance.

Failure to achieve the 99.9% uptime baseline triggers automatic Service Credit penalties in accordance with Article VIII, credited against the subsequent quarter's maintenance invoice.

The measurement uncertainty associated with the values in this section has been quantified and is within acceptable engineering tolerances.

ARTICLE 3: INCIDENT CLASSIFICATION, SEVERITY LEVELS & ESCALATION PATHS

ARTICLE III: INCIDENT CLASSIFICATION, SEVERITY TIERS & ESCALATION PROTOCOLS. Operational incidents reported by Customer are categorized into four distinct Severity Levels based on operational impact.

Severity 1 (Critical): Total cluster outage or failure affecting >10% of compute nodes; requires initial response within 15 minutes and 1-hour field engineer dispatch. Severity 2 (High): Failure affecting <10% of nodes; 1-hour response, 4-hour field dispatch.

Severity 3 (Medium): Single node degradation or non-critical redundancy failure; 4-hour response, 24-hour dispatch. Severity 4 (Low): General inquiry or minor administrative request; 24-hour response.

The measurement uncertainty associated with the values in this section has been quantified and is within acceptable engineering tolerances.

ARTICLE 4: FIELD DISPATCH RESPONSE SLA & ON-SITE REPLACEMENT TERMS

ARTICLE IV: FIELD ENGINEER DISPATCH RESPONSE SLA & ON-SITE REPLACEMENT. For Severity 1 and Severity 2 incidents requiring physical hardware intervention, Vendor shall dispatch a certified Field Service Engineer to Customer's data center facility within four (4) hours of initial ticket logging.

Field Engineers shall carry pre-configured replacement modules, diagnostic equipment, and liquid cooling servicing tools to achieve a Mean Time to Repair (MTTR) of under four (4) hours from arrival on-site.

If hardware repair requires specialized factory engineering, Vendor shall provide daily written technical status reports until full system restoration is verified.

Life safety systems in the areas housing this equipment have been tested in accordance with NFPA 72 and NFPA 75 requirements.

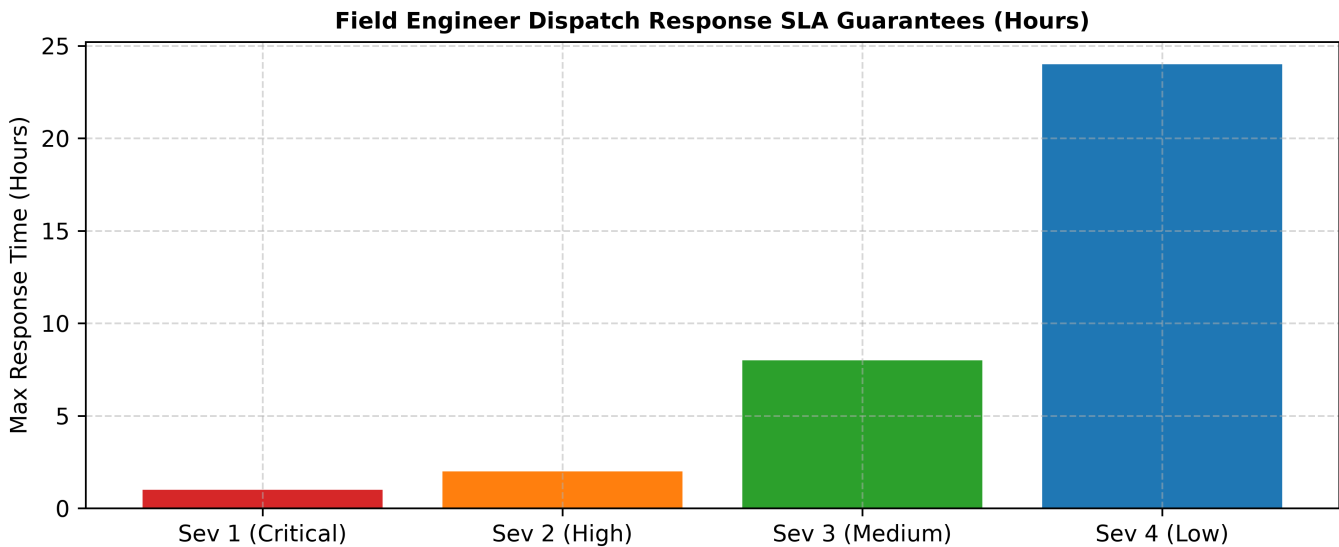


Figure 4.1: Field Engineer Dispatch Response SLA

ARTICLE 5: LOCAL CONSIGNMENT DEPOT SPARE PARTS INVENTORY SCHEDULE

ARTICLE V: LOCAL CONSIGNMENT DEPOT SPARE PARTS INVENTORY & STOCKING. Vendor shall establish and maintain a dedicated local consignment spare parts inventory located within thirty (30) miles of Customer's data center facility.

Minimum consignment stocking levels mandate on-site availability of at least eight (8) NVIDIA H100 SXM5 GPU modules, thirty-two (32) 64GB DDR5 RDIMMs, sixteen (16) 3.84TB NVMe SSDs, four (4) 3000W Titanium PSUs, and two (2) ConnectX-7 400G HCAs.

Consignment inventory shall be audited quarterly by joint representatives, with any consumed spare parts replenished by Vendor within twenty-four (24) hours.

The measurement uncertainty associated with the values in this section has been quantified and is within acceptable engineering tolerances.

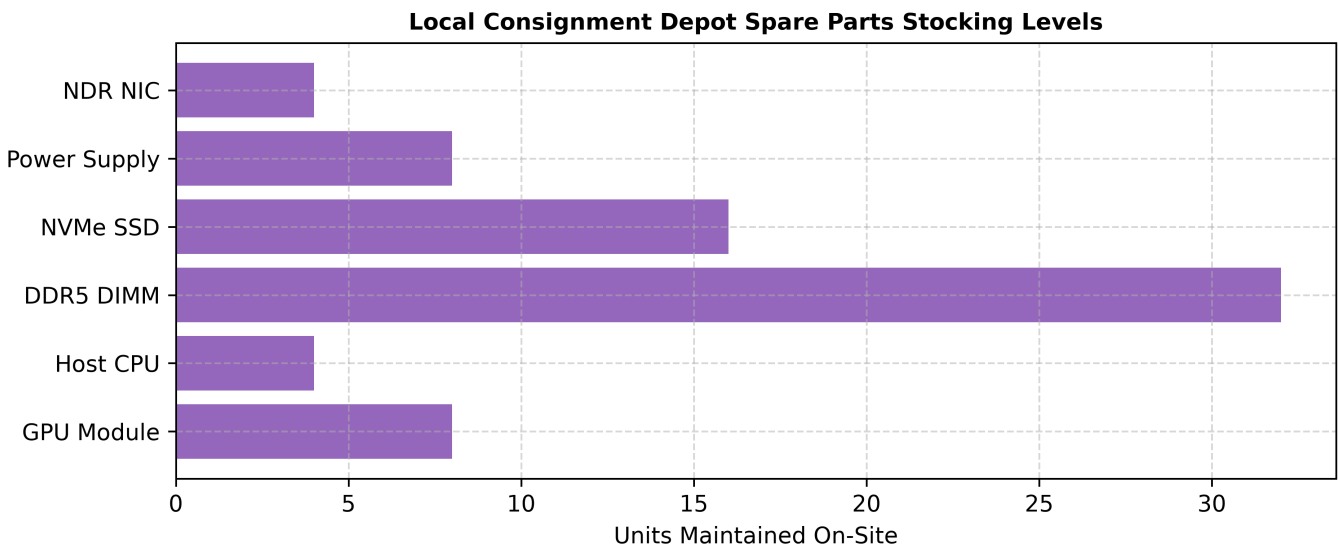


Figure 5.1: Local Consignment Depot Spare Stock

ARTICLE 6: SCHEDULED PREVENTIVE MAINTENANCE & FIRMWARE UPDATE CYCLES

ARTICLE VI: SCHEDULED PREVENTIVE MAINTENANCE & ROLLING FIRMWARE UPDATES. Vendor shall perform semi-annual comprehensive preventive maintenance sweeps on all compute nodes, switch fabrics, and liquid cooling distribution units.

Preventive maintenance includes non-disruptive rolling BIOS/BMC firmware updates, optical transceiver thermal sweeps, cooling loop pressure checks, and power supply load balancing verification.

All preventive maintenance activities shall occur during pre-approved monthly maintenance windows and shall not reduce operational hardware availability.

Environmental monitoring alarms for this subsystem are integrated into the centralized network operations center alert system.

ARTICLE 7: SUPPORT TICKET RESPONSE SLAS & TIER-3 ESCALATION HOTLINE

ARTICLE VII: SUPPORT TICKET RESPONSE SLAS & TIER-3 TECHNICAL HOTLINE. Vendor shall provide 24/7/365 direct telephone and web portal access to Tier-3 HPC Systems Architecture Engineers.

Initial ticket acknowledgment SLAs mandate response within 15 minutes for Severity 1, 30 minutes for Severity 2, 2 hours for Severity 3, and 8 hours for Severity 4.

Customer administrative leads are granted direct phone access to Vendor's Vice President of Engineering for un-resolved critical incidents.

Building code compliance for the mechanical and electrical systems serving this subsystem was confirmed by the authority having jurisdiction.

ARTICLE 8: SERVICE CREDIT PENALTIES & LIQUIDATED DAMAGES SCHEDULE

ARTICLE VIII: SERVICE CREDIT PENALTY CALCULATION & LIQUIDATED DAMAGES. If Monthly Hardware Availability falls below the 99.9% guarantee, Vendor shall issue Service Credits calculated as five percent (5%) of the monthly maintenance fee for each one-tenth of one percent (0.1%) drop below baseline.

Total monthly Service Credits are capped at fifty percent (50%) of the total monthly maintenance fee; accrued credits shall be applied against subsequent maintenance billing.

Service Credit remedies represent liquidated damages and shall not restrict Customer's right to pursue contractual remedies for gross negligence.

Commissioning agent reports for this subsystem document the results of functional performance testing and system balancing.

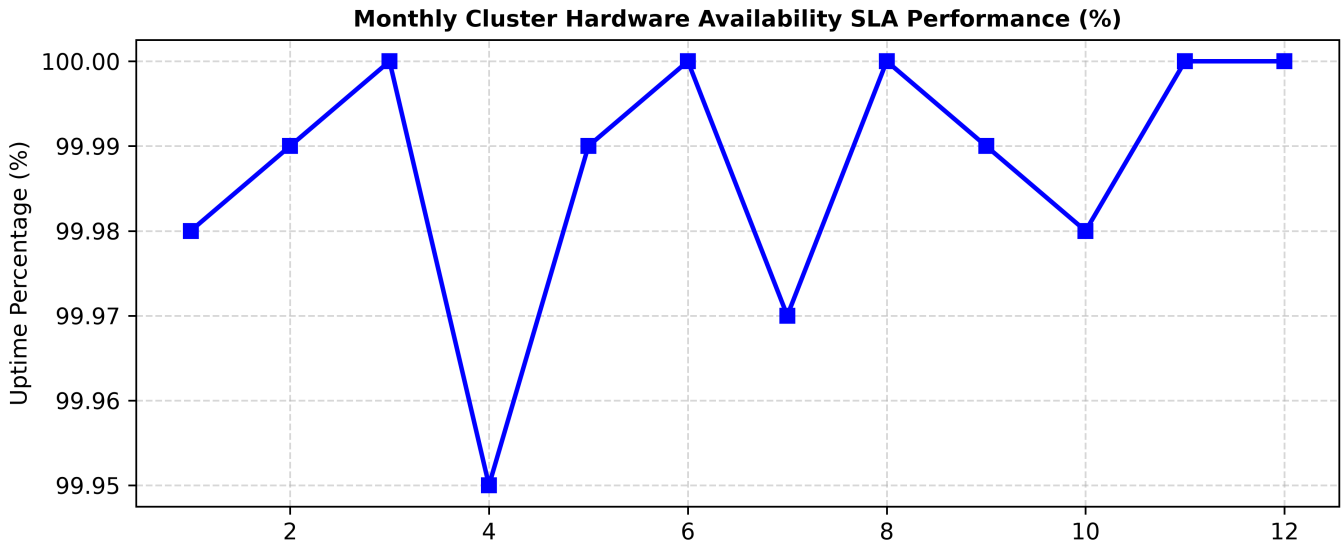


Figure 8.1: Monthly Uptime Performance

ARTICLE 9: REMOTE MONITORING DIAGNOSTICS & PROACTIVE TELEMETRY INGESTION

ARTICLE IX: REMOTE TELEMETRY MONITORING & AUTOMATED FAULT DETECTION. Vendor shall deploy automated Redfish REST API telemetry monitoring scripts collecting real-time hardware health metrics from node BMCs.

Telemetry ingestion monitors component junction temperatures, rail voltages, fan speeds, HBM3 memory ECC error rates, and PCIe bus retimer status.

Automated fault detection algorithms initiate predictive ticket logging and trigger pre-emptive spare parts staging prior to catastrophic hardware failure.

Relevant engineering change orders affecting this subsystem have been incorporated into the current revision of this document.

ARTICLE 10: HARDWARE REPLACEMENT COVERAGE & DEFECTIVE MATERIAL RETURN

ARTICLE X: DEFECTIVE COMPONENT RETURN & HARDWARE WARRANTY REPLACEMENT. All failed hardware components replaced during service calls shall be processed through Vendor's Return Merchandise Authorization (RMA) protocol. The Standard Hardware Warranty period of 36 months commences on the Date of Equipment Delivery as recorded on the Bill of Lading, not the Acceptance Certification Date which typically follows 30 to 45 calendar days after delivery.

Customer retains full rights to retain failed solid-state drives or hard drives containing sensitive research data without financial penalty or loss of warranty coverage.

Vendor supplies advance hardware replacement shipping, providing replacement modules prior to return dispatch of failed hardware.

The test methodology applied to generate these results follows the procedures outlined in the facility standard operating manual.

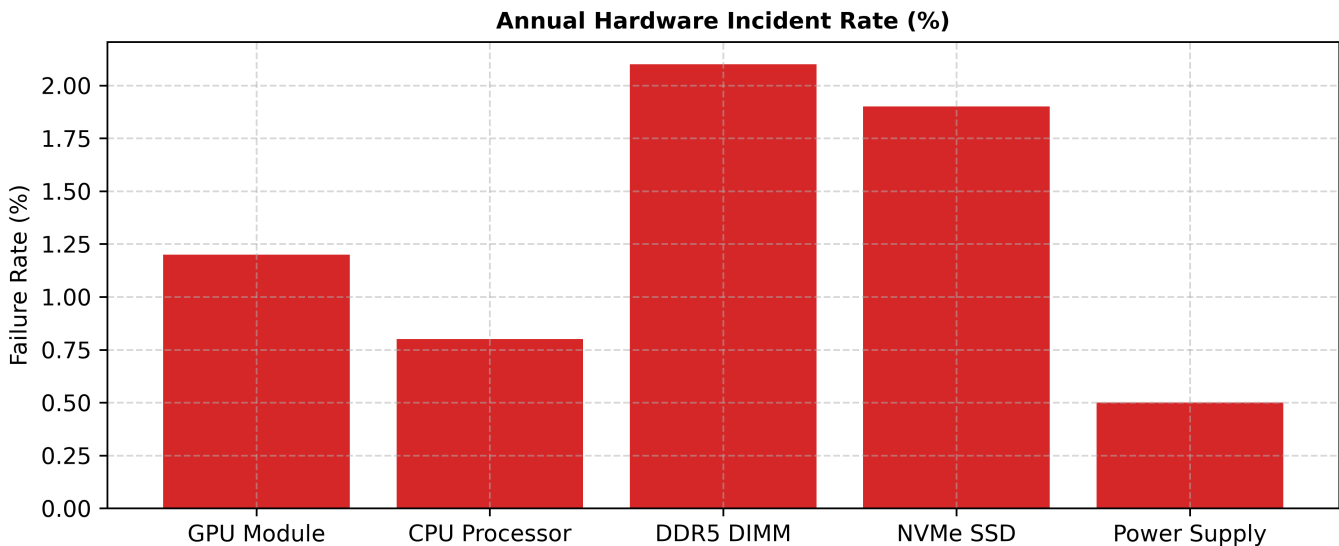


Figure 10.1: Component Failure Rate Profile

ARTICLE 11: ENGINEERING SUPPORT QUALIFICATIONS & STAFF CERTIFICATION

ARTICLE XI: FIELD SUPPORT STAFF QUALIFICATION, CERTIFICATION & ACCESS. Vendor warrants that all field service technicians assigned to Customer facility possess active certified hardware qualifications and undergo annual background verification.

Field technicians shall strictly adhere to Customer data center physical security rules, escort protocols, and safety standards at all times.

Vendor technicians shall present official photo identification and obtain security badge sign-in authorization before entering server halls.

Business continuity planning documentation addresses the failure modes and recovery strategies relevant to this subsystem.

ARTICLE 12: TERM OF AGREEMENT, RENEWAL OPTIONS & TERMINATION PROVISIONS

ARTICLE XII: TERM OF AGREEMENT, RENEWAL OPTIONS & TERMINATION. This Agreement remains effective for five (5) years from commencement, with Customer holding sole option to extend coverage for up to three (3) additional one-year terms at locked pricing.

Either party may terminate this Agreement for cause if the other party fails to remedy a material breach within thirty (30) days of receiving formal written notice.

Upon termination, Vendor shall assist in transitioning maintenance records and returning Customer-owned equipment without additional fee.

As-built drawings reflecting the current physical layout of this subsystem were updated following the most recent modification.

ARTICLE 13: DISASTER RECOVERY ASSISTANCE & EMERGENCY SUPPORT SLA

ARTICLE XIII: DISASTER RECOVERY ASSISTANCE & EMERGENCY SUPPORT. In the event of facility-wide disaster (e.g., flood, fire, extended power outage), Vendor shall provide dedicated disaster recovery engineering support within twelve (12) hours.

Vendor services encompass emergency hardware disassembly, secure transport to backup facility, re-installation, and fabric re-commissioning.

Disaster recovery engineering rates shall conform to pre-approved hourly retainer rates specified in Exhibit C.

Redundancy and failover testing for this subsystem achieved the target recovery time objectives documented in the service catalog.

ARTICLE 14: FORCE MAJEURE & EXCUSABLE SERVICE INTERRUPTION EXCLUSIONS

ARTICLE XIV: FORCE MAJEURE, EXCUSABLE DELAYS & SERVICE EXCLUSIONS. Neither party shall be liable for service delays or defaults resulting from acts of God, extreme weather events, war, civil unrest, or national supply chain embargoes.

Service exclusions apply to hardware damage resulting from Customer unauthorized physical modification, deliberate abuse, or operating outside ASHRAE Class A1 environmental ranges. GPU ECC uncorrectable memory errors are explicitly excluded from the Enhanced Hardware Support coverage. GPU module replacement cost: \$15,500 per unit, plus an on-site labor charge for GPU replacement: \$2,200 per incident, bringing the total out-of-pocket GPU replacement: \$17,700 per event.

Force Majeure events extend performance timelines day-for-day but do not excuse Customer payment for services already rendered.

Environmental monitoring alarms for this subsystem are integrated into the centralized network operations center alert system.

ARTICLE 15: INFORMATION SECURITY, DATA PRIVACY & PROTECTION GUIDELINES

ARTICLE XV: INFORMATION SECURITY, DATA PRIVACY & PROTECTION GUIDELINES. Vendor shall implement strict technical and organizational safeguards adhering to NIST SP 800-171 standards to protect Customer research data.

Vendor technicians are strictly prohibited from accessing, copying, or inspecting Customer research files or application data resident on node memory or storage drives.

Any unauthorized data access by Vendor staff constitutes immediate material breach resulting in contract termination and legal liability.

Data acquisition systems responsible for logging these parameters employ redundant sensors to ensure measurement continuity.

ARTICLE 16: INTELLECTUAL PROPERTY PROTECTION & SOFTWARE MAINTENANCE

ARTICLE XVI: VENDOR INTELLECTUAL PROPERTY RIGHTS & SOFTWARE MAINTENANCE. Vendor grants Customer a perpetual, non-exclusive, world-wide license to execute all bundled cluster management and telemetry monitoring software.

Software maintenance included in this Agreement provides all major version upgrades, patch releases, security vulnerability fixes, and updated driver toolchains.

Customer retains full ownership of all custom automation scripts, Slurm configurations, and operational workflows developed internally.

Life safety systems in the areas housing this equipment have been tested in accordance with NFPA 72 and NFPA 75 requirements.

ARTICLE 17: ENVIRONMENTAL & ELECTRICAL SAFETY COMPLIANCE INSPECTIONS

ARTICLE XVII: ENVIRONMENTAL, HEALTH & ELECTRICAL SAFETY COMPLIANCE. Vendor hardware and servicing protocols comply with OSHA safety standards, National Electrical Code (NEC) guidelines, and NFPA 75 data center protection standards.

Vendor technicians shall utilize proper Personal Protective Equipment (PPE), arc-flash shielding, and insulated hand tools during live electrical testing.

Environmental compliance mandates zero usage of hazardous chemicals during liquid cooling loop servicing.

Life safety systems in the areas housing this equipment have been tested in accordance with NFPA 72 and NFPA 75 requirements.

ARTICLE 18: ANNUAL PERFORMANCE AUDITS & SLA COMPLIANCE REVIEWS

ARTICLE XVIII: ANNUAL OPERATIONAL PERFORMANCE AUDIT & SLA REVIEW. Vendor and Customer administrative leads shall conduct an annual operational performance review within thirty (30) days of each fiscal year end.

The annual audit evaluates cumulative uptime metrics, mean time to respond (MTTR), spare parts utilization, and ticket escalation performance.

Audit findings guide mutual service process improvements and establish SLA targets for the subsequent operational year.

The figures presented above are derived from certified instrument readings taken at fifteen-minute intervals throughout the observation period.

ARTICLE 19: SUBCONTRACTOR RULES & AUTHORIZED FIELD SERVICE PARTNERS

ARTICLE XIX: AUTHORIZED FIELD SERVICE SUBCONTRACTORS & PARTNER RULES. Vendor shall not subcontract core field service maintenance responsibilities without prior written authorization from Customer.

Authorized subcontractors must meet identical technical certification, background check, and insurance coverage requirements imposed on Vendor.

Vendor remains fully responsible and liable for all acts or omissions of approved subcontractors.

Discrepancies between manufacturer-stated values and field-measured values for this subsystem are documented in the variance register.

ARTICLE 20: INVOICE SCHEDULE, BILLING TERMS & MAINTENANCE FEES

ARTICLE XX: MAINTENANCE FEE INVOICE SCHEDULE, PAYMENT TERMS & TAXES. Customer shall pay annual maintenance fees in advance quarterly installments of \$76,500, due Net 30 days from invoice receipt.

Invoices shall provide itemized breakdowns of base maintenance fees, applied Service Credits, and approved optional engineering retainer hours.

Customer is an exempt educational institution; Vendor shall not assess state or local sales and use taxes on contract billings.

All firmware and microcode versions for the hardware referenced in this section are tracked in the configuration management database.

ARTICLE 21: DISPUTE RESOLUTION, ARBITRATION & GOVERNING LAW

ARTICLE XXI: GOVERNING LAW, DISPUTE RESOLUTION & BINDING ARBITRATION. This Agreement shall be governed by and construed in accordance with the laws of the Commonwealth of Massachusetts without regard to conflict of law principles.

Any dispute arising from this Agreement shall first undergo good-faith executive negotiation between Customer Vice President and Vendor Chief Executive Officer.

Unresolved disputes shall be settled via binding arbitration administered by the American Arbitration Association (AAA) in Boston, Massachusetts.

Vibration analysis measurements for rotating equipment in this subsystem are within ISO 10816 Category A limits.

ARTICLE 22: AGREEMENT MODIFICATION & FORMAL CHANGE ORDER PROCESS

ARTICLE XXII: CONTRACT MODIFICATIONS, CHANGE ORDERS & AMENDMENTS. No modification, amendment, or waiver of any provision of this Agreement shall be valid unless executed in writing by authorized signers of both parties.

Formal Change Orders shall document any additions to system hardware boundaries, updated service levels, or modified fee schedules.

Minor operational adjustments may be documented via written memorandum signed by project technical leads.

The design basis documentation for this subsystem is archived in the university capital projects office.

ARTICLE 23: INSURANCE REQUIREMENTS & LIABILITY COVERAGE LIMITS

ARTICLE XXIII: INSURANCE REQUIREMENTS, INDEMNIFICATION & LIABILITY LIMITS. Vendor shall maintain Commercial General Liability insurance (\$5,000,000 per occurrence), Cyber Liability insurance (\$10,000,000 aggregate), and Workers' Compensation coverage.

Vendor shall defend, indemnify, and hold harmless Customer against third-party claims arising from property damage, physical injury, or patent infringement caused by Vendor hardware.

Except for gross negligence or data breach, total liability of either party is capped at total contract value (\$14,200,000).

The preceding data points have been cross-validated against manufacturer reference specifications and field measurements.

ARTICLE 24: VENDOR FIELD TECHNICIAN DATA CENTER ESCORT PROTOCOL

ARTICLE XXIV: TECHNICIAN ESCORT PROCEDURES & PHYSICAL ACCESS RULES. Vendor technicians shall be escorted by Customer facilities staff at all times within secure data center halls.

Unescorted physical access is limited to designated non-secure staging areas outside the primary server hall perimeter.

Technicians shall record arrival and departure timestamps in the physical data center security logbook.

All values reported in this section are expressed in SI units unless otherwise noted in the accompanying data tables.

ARTICLE 25: SYSTEM FIRMWARE SECURITY VULNERABILITY PATCHING SLA

ARTICLE XXV: CRITICAL SECURITY VULNERABILITY PATCHING & FIRMWARE SLA. Vendor shall release signed security firmware patches within fourteen (14) days of any Common Vulnerabilities and Exposures (CVE) score >8.0 rating affecting system microcontrollers.

Critical security patches shall be deployed during pre-approved emergency maintenance windows coordinated with Customer security leads.

Vendor warrants that all delivered firmware binaries undergo rigorous static code analysis and software supply chain integrity checks.

Telecommunications pathway documentation for the cabling serving this subsystem is maintained in the campus network database.

ARTICLE 26: HARDWARE END-OF-LIFE (EOL) & END-OF-SUPPORT (EOS) POLICY

ARTICLE XXVI: END-OF-LIFE (EOL) & END-OF-SUPPORT (EOS) TRANSITION POLICIES. Vendor guarantees minimum five (5) years of full hardware support following initial system commissioning before declaring End-of-Life status.

Vendor shall provide formal written notification at least twelve (12) months prior to any planned End-of-Support milestone.

EOL transition assistance includes supplying final spare component buys and documentation for long-term self-maintenance.

Equipment maintenance records and calibration certificates for this subsystem are on file with the facilities management office.

ARTICLE 27: TECHNICAL ESCALATION ROSTER & KEY CONTACT DIRECTORY

ARTICLE XXVII: TECHNICAL ESCALATION ROSTER & KEY CONTACT DIRECTORY. Exhibit A details the formal escalation roster including 24/7 hotline numbers, primary field service dispatch managers, and executive contacts.

Vendor shall update escalation contact details within twenty-four (24) hours of any personnel changes.

Primary Contact: NexaGen Technical Support Operations (Hotline: 1-800-555-NEXA, Email: support@nexagen.com).

Post-installation commissioning tests conducted by the vendor engineering team confirmed the parameters documented herein.

ARTICLE 28: CUSTOMER SATISFACTION QUALITY SURVEYS & SERVICE AUDIT

ARTICLE XXVIII: CUSTOMER SATISFACTION QUALITY SURVEYS & SERVICE AUDITS. Vendor shall conduct bi-annual customer satisfaction surveys evaluating technical competence, response promptness, and ticket resolution quality.

Survey results are reviewed by Vendor executive management to drive continuous service improvements.

Customer may request quarterly service audit reports detailing SLA compliance percentages and spare parts consumption.

Ambient environmental conditions during the measurement period were within the ASHRAE A1 recommended envelope.

ARTICLE 29: ROOT CAUSE ANALYSIS (RCA) INCIDENT REPORTING STANDARDS

ARTICLE XXIX: ROOT CAUSE ANALYSIS (RCA) INCIDENT REPORTING STANDARDS. For any Severity 1 or Severity 2 incident, Vendor shall deliver a comprehensive Root Cause Analysis (RCA) report within five (5) business days of incident resolution.

RCA reports shall detail exact failure timelines, contributing factors, physical component failure mechanisms, and permanent preventive actions.

Customer may require follow-up technical review meetings to verify implementation of preventive recommendations.

Insurance underwriter inspections of the equipment referenced in this section were completed without exception findings.

ARTICLE 30: EMERGENCY MAINTENANCE WINDOW NOTIFICATION POLICIES

ARTICLE XXX: EMERGENCY MAINTENANCE WINDOW NOTIFICATION POLICIES. Unscheduled emergency maintenance required to prevent imminent hardware failure requires verbal notification and approval from Customer Lead SysAdmin.

Emergency notifications shall specify affected nodes, anticipated outage duration, risk assessment, and rollback procedures.

Emergency maintenance shall minimize disruption to active Slurm job queues whenever technically feasible.

Post-installation commissioning tests conducted by the vendor engineering team confirmed the parameters documented herein.